# Math 271: Mathematical Methods (Semester I, 2026–27)

Class Test 5 (28 September 2026) — Model Solution and Exposition

## 0. Set-up

**Notation.** Throughout, $\|\cdot\|$ is the Euclidean norm and $B_r(a) = \{x : \|x - a\| < r\}$ is the open ball of radius $r$ about $a$. We are given a $C^1$ map $f : \mathbb{R}^n \to \mathbb{R}^m$ with $f(x^*) = b^*$; $Df(x^*)$ is the $m \times n$ Jacobian of $f$ at $x^*$. Recall the two definitions of the test:

- $f$ is *locally onto* at $x^*$ if there are $r, s > 0$ such that *for every* $b \in B_s(b^*)$ *there is at least one* $x \in B_r(x^*)$ with $f(x) = b$;
- $f$ is *locally one-to-one* at $x^*$ if there are $r, s > 0$ such that *for every* $b \in B_s(b^*)$ *there is at most one* $x \in B_r(x^*)$ with $f(x) = b$; equivalently,

$$x, x' \in B_r(x^*), \quad f(x) = f(x') \in B_s(b^*) \implies x = x'.$$

As we will rely heavily on the Implicit Function Theorem for proving the results asked in Class Test 5, I am quoting it here for reference. (To avoid a clash with the $x^*$ and $y^*$ of the test, the endogenous variables are called $y$ and the exogenous variables $z$.)

> **Implicit Function Theorem (IFT).** *Let $G : \mathbb{R}^k \times \mathbb{R}^p \to \mathbb{R}^k$ be $C^1$ with $G(y^\circ, z^\circ) = 0$, and suppose that the $k \times k$ matrix $D_y G(y^\circ, z^\circ)$ is nonsingular. Then there are balls $W := B_\rho(y^\circ) \subset \mathbb{R}^k$ and $B := B_\sigma(z^\circ) \subset \mathbb{R}^p$ and a $C^1$ map $\phi : B \to W$ such that $\phi(z^\circ) = y^\circ$ and $G(\phi(z), z) = 0$ for all $z \in B$. Moreover (**uniqueness clause**), for $z \in B$, $\phi(z)$ is the only $y \in W$ with $G(y, z) = 0$:*
>
> $$\forall\, z \in B \;\; \forall\, y \in W : \qquad G(y, z) = 0 \implies y = \phi(z).$$

Two things must be checked every time the theorem is used: the system is *square* (as many equations as endogenous variables) and the Jacobian with respect to the endogenous variables is *nonsingular at the base point*. We also use the Chain Rule.

**Renumbering.** Permuting the coordinates of $\mathbb{R}^n$ (or of $\mathbb{R}^m$) does not change Euclidean distances, so it maps balls to balls with the same centre and radius, and it preserves the $C^1$ property. So we may renumber variables or equations without loss of generality.

## 1. Part (a)(i): $\operatorname{rank} Df(x^*) = m \le n \;\Rightarrow\; f$ is locally onto at $x^*$

**Claim.**

$$\operatorname{rank} Df(x^*) = m \le n \implies \exists\, r, s > 0 \;\; \forall\, b \in B_s(b^*) \;\; \exists\, x \in B_r(x^*) : \;\; f(x) = b.$$

**Idea.** This is the nonlinear version of solving $Ax = b$ when $\operatorname{rank} A = m$: pick $m$ "basic" variables whose columns form a nonsingular block, fix the remaining "free" variables, and solve for the basic ones. The right-hand side $b$ is treated as an exogenous variable.

**Step 1 (choose the endogenous variables).** Row rank = column rank = $m$, so $Df(x^*)$ has $m$ linearly independent *columns*. Renumber the variables so that these are the first $m$; then the $m \times m$ block

$$D_u f(x^*) = \Big(\frac{\partial f_i}{\partial x_j}(x^*)\Big)_{1 \le i,j \le m}$$

is nonsingular. Write $x = (u, w)$ with $u = (x_1, \ldots, x_m) \in \mathbb{R}^m$, $w = (x_{m+1}, \ldots, x_n) \in \mathbb{R}^{n-m}$, and $x^* = (u^*, w^*)$. (If $m = n$ there is no $w$, and everything below goes through with $w$ deleted.)

**Step 2 (set up a square system).** Define $G : \mathbb{R}^m \times (\mathbb{R}^{n-m} \times \mathbb{R}^m) \to \mathbb{R}^m$ by

$$G(u;\, w, b) := f(u, w) - b,$$

with $u$ endogenous ($m$ unknowns) and $(w, b)$ exogenous. This is $m$ equations in $m$ unknowns. $G$ is $C^1$ because $f$ is, and at the point $(u^*; w^*, b^*)$:

$$G(u^*; w^*, b^*) = f(x^*) - b^* = 0, \qquad D_u G(u^*; w^*, b^*) = D_u f(x^*) \text{ is nonsingular.}$$

**Step 3 (apply the IFT).** There are balls $W = B_\rho(u^*) \subset \mathbb{R}^m$, $B = B_\sigma(w^*, b^*) \subset \mathbb{R}^{n-m} \times \mathbb{R}^m$ and a $C^1$ (in particular continuous) map $\phi : B \to W$ with

$$\phi(w^*, b^*) = u^* \qquad \text{and} \qquad f(\phi(w, b), w) = b \quad \text{for all } (w, b) \in B.$$

**Step 4 (freeze the free variables and place the preimage in the ball).** Freeze $w = w^*$ (only $w$; $b$ stays free). Note $\|(w^*, b) - (w^*, b^*)\| = \|b - b^*\|$, so $(w^*, b) \in B$ whenever $\|b - b^*\| < \sigma$. Now let $r > 0$ be arbitrary. The map $b \mapsto \phi(w^*, b)$ is continuous at $b^*$ with value $u^*$, so there is $s$ with $0 < s \le \sigma$ and

$$\|b - b^*\| < s \implies \|\phi(w^*, b) - u^*\| < r.$$

For $b \in B_s(b^*)$ put

$$x_b := (\phi(w^*, b),\, w^*) \in \mathbb{R}^n.$$

Then $f(x_b) = f(\phi(w^*, b), w^*) = b$ by Step 3, and, since the $w$-components of $x_b$ and $x^*$ agree,

$$\|x_b - x^*\| = \|\phi(w^*, b) - u^*\| < r.$$

**Conclusion.** For every $b \in B_s(b^*)$ the point $x_b \in B_r(x^*)$ satisfies $f(x_b) = b$. Hence $f$ is locally onto at $x^*$. ■

**Remark.** (1) We proved slightly more than asked: for *every* $r > 0$ there is a suitable $s > 0$. (2) One can avoid the continuity argument by taking $r := \rho$ (the radius of $W$) and $s := \sigma$: then $\phi(w^*, b) \in W$ gives $\|x_b - x^*\| = \|\phi(w^*, b) - u^*\| < \rho$ directly. (3) Equivalently one may freeze $w$ *before* applying the theorem: $g(u) := f(u, w^*)$ is a $C^1$ map $\mathbb{R}^m \to \mathbb{R}^m$ with $Dg(u^*) = D_u f(x^*)$ nonsingular, and the IFT for $g(u) - b = 0$ gives the same conclusion. What matters is that the frozen value is $w^*$, so that the base point is $x^*$. (4) Step 1 cannot be replaced by "$m$ independent rows": every one of the $m$ rows is needed, and the nonsingular block must be a block of *columns*, i.e. it decides which *variables* are endogenous.

**Example** ($n = 2$, $m = 1$)**.** $f(u, w) = u + w^2$ at $x^* = (0, 0)$, $b^* = 0$. Here $Df(x^*) = (1 \;\; 0)$ has rank 1, and the nonsingular $1 \times 1$ block is the $u$-column. The level set $\{f = b\}$ is the parabola $u = b - w^2$, i.e. $\phi(w, b) = b - w^2$. Freezing $w = w^* = 0$ gives $x_b = (b, 0)$: as $b$ runs through $(-s, s)$, $x_b$ runs along a segment through $x^*$, and it stays in $B_r(x^*)$ when $s \le r$. Each level curve near $\{f = 0\}$ meets the frozen line exactly once, inside the ball. (Freezing the *other* variable would fail: for $u = 0$ the equation $w^2 = b$ has no solution when $b < 0$. This is why the block must be chosen by the rank condition.)

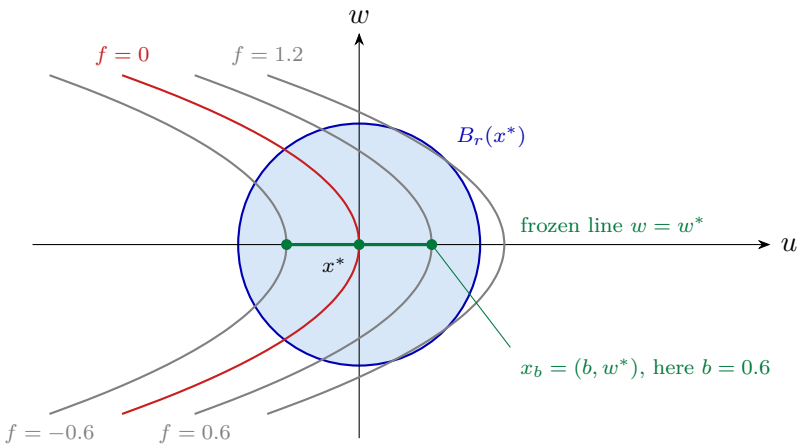



Figure 1: Part (a)(i) for $f(u,w) = u + w^2$. Grey/red curves: level sets $\{f = b\}$. Freezing the free variable $w = w^*$ and solving for the endogenous $u$ produces, for every $b$ near $b^*$, one point $x_b$ on the frozen line with $f(x_b) = b$; continuity of $\phi$ keeps $x_b$ inside $B_r(x^*)$.

## 2. Part (a)(ii): $\operatorname{rank} Df(x^*) = n \le m \;\Rightarrow\; f$ is locally one-to-one at $x^*$

**Claim.**

$$\operatorname{rank} Df(x^*) = n \le m \implies \exists\, r, s > 0 : \quad \big(x, x' \in B_r(x^*),\ f(x) = f(x') \in B_s(b^*)\big) \Rightarrow x = x'.$$

**Idea.** This is the nonlinear version of: if $\operatorname{rank} A = n$ then some $n$ of the equations $Ax = b$ already determine $x$. Now the nonsingular block is a block of *rows*.

**Step 1 (choose $n$ equations).** Row rank $= n$, so $Df(x^*)$ has $n$ linearly independent *rows*. Renumber the equations (components of $f$) so that these are the first $n$, and put

$$f_I := (f_1, \dots, f_n) : \mathbb{R}^n \to \mathbb{R}^n, \qquad b_I := (b_1, \dots, b_n) \ \text{ for } b \in \mathbb{R}^m.$$

Then $Df_I(x^*)$, the $n \times n$ matrix formed by the first $n$ rows of $Df(x^*)$, is nonsingular.

**Step 2 (apply the IFT and keep its uniqueness clause).** Let $c^* := f_I(x^*) = b_I^*$ and consider the square system

$$G(x, c) := f_I(x) - c = 0, \qquad x \in \mathbb{R}^n \text{ endogenous}, \quad c \in \mathbb{R}^n \text{ exogenous}.$$

$G$ is $C^1$, $G(x^*, c^*) = 0$ and $D_x G(x^*, c^*) = Df_I(x^*)$ is nonsingular. The IFT gives balls $W = B_\rho(x^*)$, $B = B_\sigma(c^*)$, a $C^1$ map $\phi : B \to W$ and

$$\forall\, c \in B \ \ \forall\, x \in W : \qquad f_I(x) = c \implies x = \phi(c). \tag{U}$$

**Step 3 (choose the radii).** To use (U) on two points we need *both* points in $W$ *and* their common value in $B$. Since $f_I$ is continuous at $x^*$ and $f_I(x^*) = c^*$, there is $r$ with $0 < r \le \rho$ and

$$f_I\big(B_r(x^*)\big) \subset B.$$

Let $s > 0$ be arbitrary (e.g. $s = 1$).

**Step 4 (conclude).** Let $b \in B_s(b^*)$ and $x, x' \in B_r(x^*)$ with $f(x) = f(x') = b$. Looking at the first $n$ components, $f_I(x) = f_I(x') = b_I =: c$. Now $x \in B_r(x^*) \subset W$ and $c = f_I(x) \in B$ by Step 3, and likewise for $x'$. By (U), $x = \phi(c) = x'$.

**Conclusion.** Every $b \in B_s(b^*)$ has at most one preimage in $B_r(x^*)$, i.e. $f$ is locally one-to-one at $x^*$. ∎

**Remark.** (1) Instead of shrinking the $x$-ball one may keep $B_r(x^*) = W$ and shrink the $b$-ball: take $0 < s \le \sigma$. Then $b \in B_s(b^*)$ gives $\|b_I - b_I^*\| \le \|b - b^*\| < \sigma$ (dropping coordinates does not increase distance), so $c = b_I \in B$ and (U) applies. (2) Without either precaution the argument has a gap: two points of $W$ may have a common value outside $B$, about which (U) says nothing; see the example below. (3) The *uniqueness* clause is essential. Existence of $\phi$, or "$\phi$ is the unique continuous solution through $x^*$", does not by itself rule out a second solution $x' \in B_r(x^*)$ of $f(x') = b$.

**Example** (why one must shrink; $n = m = 1$)**.** $g(x) = x^3 - 3x$ at $x^* = 0$, $g'(0) = -3 \neq 0$. The pair $W = (-1.2, 1.2)$, $B = (-1, 1)$ satisfies the uniqueness clause: for $c \in (-1, 1)$ there is exactly one root of $g(x) = c$ in $(-1, 1)$, and none with $1 \le |x| < 1.2$ because there $1.872 < |g(x)| \le 2$. But $W$ itself is *not* a ball on which $g$ is one-to-one: $-1.1$ and $\approx -0.8965$ both lie in $W$ and $g(-1.1) = g(-0.8965) \approx 1.969$, a value outside $B$. Shrinking to $B_r(0)$ with $g(B_r(0)) \subset B$ (e.g. $r = 0.3$, since $|g(x)| \le 3|x| + |x|^3 < 1$ for $|x| < 0.3$) removes the problem.

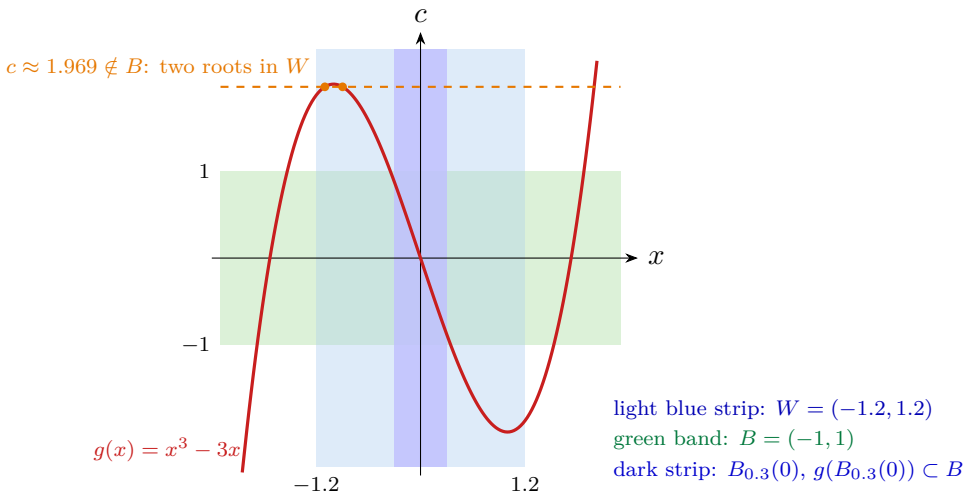



Figure 2: Part (a)(ii) in one variable. The uniqueness clause only speaks about values $c$ in the green band $B$. The two orange points lie in $W$ and share a value outside $B$, so $g$ is not one-to-one on $W$. On the shrunken ball $B_{0.3}(0)$ every value lies in $B$, and the clause makes $g$ one-to-one there.

## 3. Part (b): the Inverse Function Theorem

**Claim.** Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be $C^1$ with $f(x^*) = y^*$ and $Df(x^*)$ nonsingular. Then there are a ball $B_r(x^*)$ and an open set $V \ni y^*$ such that $f : B_r(x^*) \to V$ is one-to-one and onto, the inverse $f^{-1} : V \to B_r(x^*)$ is $C^1$, and $Df^{-1}(f(x^*)) = (Df(x^*))^{-1}$.

**Idea.** Since $Df(x^*)$ is $n \times n$ and nonsingular, its rank is both $m$ and $n$, so (a)(i) and (a)(ii) both apply. But they give *different* balls, no set $V$, and no information about smoothness of the inverse. So we apply the IFT once, to $f(x) - y = 0$ with $x$ endogenous and $y$ exogenous: the implicit function $\phi(y)$ *is* the local inverse, and it is automatically $C^1$. The only care needed is in choosing the ball $B_r(x^*)$ and the set $V$ so that they match.

**Step 1 (apply the IFT).** Let $G(x, y) := f(x) - y$, $G : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$, with $x$ endogenous and $y$ exogenous ($n$ equations, $n$ unknowns). $G$ is $C^1$, $G(x^*, y^*) = 0$, and $D_x G(x^*, y^*) = Df(x^*)$ is nonsingular. The IFT gives balls $B_{r_0}(x^*)$, $B_{s_0}(y^*)$ and a $C^1$ map $\phi : B_{s_0}(y^*) \to B_{r_0}(x^*)$ with

$$\phi(y^*) = x^*, \qquad f(\phi(y)) = y \ \text{ for } y \in B_{s_0}(y^*),$$

$$\forall\, y \in B_{s_0}(y^*) \ \ \forall\, x \in B_{r_0}(x^*) : \qquad f(x) = y \implies x = \phi(y). \tag{U}$$

**Step 2 (choose the ball and define $V$).** To show that $f$ is one-to-one on the ball and that its inverse is the IFT map $\phi$ (Steps 3–5), we will use the **uniqueness clause** (U) of the Implicit Function Theorem. However, (U) only speaks about a pair $(x, y)$ when *both* $x \in B_{r_0}(x^*)$ *and* $y \in B_{s_0}(y^*)$.

So we need to choose a ball $B_r(x^*)$ with the following two properties:

(R1) $B_r(x^*) \subset B_{r_0}(x^*)$ (the points lie in the $x$-ball of the IFT), and

(R2) $f(x) \in B_{s_0}(y^*)$ for every $x \in B_r(x^*)$ (their values lie in the $y$-ball of the IFT).

The IFT ball $B_{r_0}(x^*)$ itself satisfies (R1), but in general *not* (R2): the IFT controls $\phi$ on $B_{s_0}(y^*)$, not $f$ on $B_{r_0}(x^*)$, and $f$ may carry points of $B_{r_0}(x^*)$ outside $B_{s_0}(y^*)$ (in the cubic example of Section 2, $-1.1 \in W = (-1.2, 1.2)$ but $g(-1.1) \approx 1.969 \notin B = (-1, 1)$).

Since $f$ is continuous at $x^*$ and $f(x^*) = y^*$, we apply the definition of continuity with $\varepsilon = s_0$: there is $\delta > 0$ such that

$$\forall\, x \in \mathbb{R}^n : \qquad \|x - x^*\| < \delta \implies \|f(x) - y^*\| < s_0.$$

Put $r := \min\{\delta, r_0\} > 0$. Then $r \le r_0$ gives (R1), $B_r(x^*) \subset B_{r_0}(x^*)$, and $r \le \delta$ gives (R2), $f(B_r(x^*)) \subset B_{s_0}(y^*)$. (If $f(B_{r_0}(x^*)) \subset B_{s_0}(y^*)$ happens to hold already, $r = r_0$ will do; the point is that this has to be checked, not assumed.)

Define

$$V := f\big(B_r(x^*)\big) = \{\, f(x) : x \in B_r(x^*) \,\}.$$

We record what this choice gives at once, and what it does not yet give.

- $y^* = f(x^*) \in V$, since $x^* \in B_r(x^*)$.
- $V \subset B_{s_0}(y^*)$ by (R2). Hence $\phi$ is defined at every point of $V$, and (U) may be applied to any $y \in V$ together with any $x \in B_r(x^*)$ (which lies in $B_{r_0}(x^*)$ by (R1)).
- $f : B_r(x^*) \to V$ is *onto by construction*: every $y \in V$ is, by definition, of the form $f(x)$ with $x \in B_r(x^*)$.
- Not yet known: that $f$ is one-to-one on $B_r(x^*)$ (Step 3) and that $V$ is open (Step 4).

**Step 3 ($f$ is one-to-one on $B_r(x^*)$).** Let $x, x' \in B_r(x^*)$ with $f(x) = f(x') =: y$. Then $y \in V \subset B_{s_0}(y^*)$ and $x, x' \in B_{r_0}(x^*)$, so (U) gives $x = \phi(y) = x'$.

**Step 4 ($V$ is open).** We show

$$V = B_{s_0}(y^*) \cap \phi^{-1}\big(B_r(x^*)\big).$$

($\subset$) Let $y \in V$, say $y = f(x)$ with $x \in B_r(x^*)$. Then $y \in B_{s_0}(y^*)$ (Step 2), and by (U), $\phi(y) = x \in B_r(x^*)$.
($\supset$) Let $y \in B_{s_0}(y^*)$ with $\phi(y) \in B_r(x^*)$. Then $y = f(\phi(y)) \in f(B_r(x^*)) = V$.
Since $\phi$ is continuous on the open ball $B_{s_0}(y^*)$, the preimage $\phi^{-1}(B_r(x^*))$ of the open ball $B_r(x^*)$ is open in $B_{s_0}(y^*)$, hence open in $\mathbb{R}^n$; so $V$ is open.

**Step 5 (bijection and inverse).** $f : B_r(x^*) \to V$ is onto by the definition of $V$ and one-to-one by Step 3. Its inverse is $\phi|_V$: for $y \in V$ we have $\phi(y) \in B_r(x^*)$ (Step 4, "$\subset$") and $f(\phi(y)) = y$; for $x \in B_r(x^*)$ we have $\phi(f(x)) = x$ by (U). Thus $f^{-1} = \phi|_V : V \to B_r(x^*)$.

**Step 6 ($f^{-1}$ is $C^1$).** $\phi$ is $C^1$ on $B_{s_0}(y^*)$ by the IFT and $V$ is an open subset, so $f^{-1} = \phi|_V$ is $C^1$ on $V$.

**Step 7 (the derivative).** For all $y \in V$ we have $f(f^{-1}(y)) = y$. Both $f$ and $f^{-1}$ are differentiable (Step 6), so the Chain Rule gives

$$Df\big(f^{-1}(y)\big)\, Df^{-1}(y) = I \qquad (y \in V).$$

At $y = y^* = f(x^*)$, where $f^{-1}(y^*) = x^*$: $Df(x^*)\, Df^{-1}(f(x^*)) = I$. Since $Df(x^*)$ is nonsingular,

$$Df^{-1}\big(f(x^*)\big) = \big(Df(x^*)\big)^{-1}.$$

■

**Remark.** (1) The order matters: we know $f^{-1}$ is differentiable *because it equals the IFT map* $\phi$; only then may the Chain Rule be applied to $f \circ f^{-1} = \mathrm{id}$. Writing "$Df^{-1}\, Df = I$ by definition of the inverse" presupposes the differentiability that is to be proved. (2) In general $f$ is *not* a bijection between the two IFT balls $B_{r_0}(x^*)$ and $B_{s_0}(y^*)$ (in the cubic example $g$ is not one-to-one on $W$); that is why we shrink to $B_r(x^*)$ and define $V$ as the image $f(B_r(x^*))$. This $V$ is open but in general *not* a ball centred at $y^*$: for $f(x) = x^2$ at $x^* = 1$ and $r = 0.4$, $V = f\big((0.6, 1.4)\big) = (0.36, 1.96)$, which is not centred at $y^* = 1$. This is why the claim asks for an open *set* $V$.

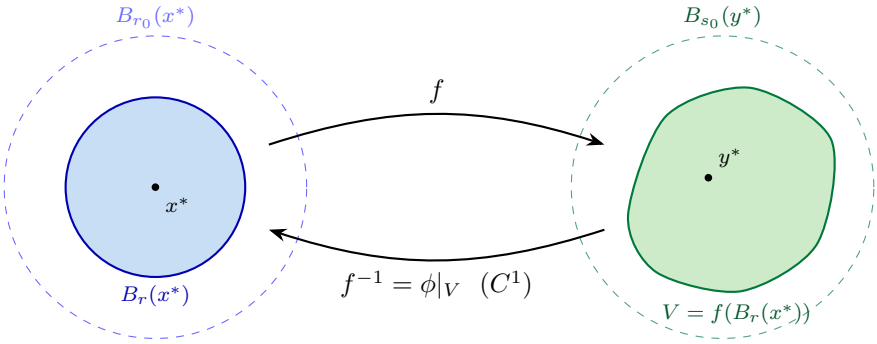



Figure 3: Part (b). The IFT supplies $\phi : B_{s_0}(y^*) \to B_{r_0}(x^*)$ (dashed balls). Shrinking to $B_r(x^*)$ with $f(B_r(x^*)) \subset B_{s_0}(y^*)$ and setting $V = f(B_r(x^*))$ makes $f : B_r(x^*) \to V$ a bijection with inverse $\phi|_V$; $V = B_{s_0}(y^*) \cap \phi^{-1}(B_r(x^*))$ is open, though in general not a ball.